
CESC 510 Course Info for Fall 2026

1 General Info

- Class meeting time: MWF 4:00 to 4:50 pm
- Room: CoAS 203
- Instructor: Dr. Jianhua Liu, Associate Professor of Electrical and Computer Engineering
- Instructor's office: LB 349
- Instructor's email: jianhua.liu@erau.edu or liu620@erau.edu
- Instructor's phone number: 226-7713 (Work)
- Instructor's office hours: MWF 1:30 pm to 3:50 pm; appointment on Zoom.
- TAs' office hours: TBD

2 Prerequisites and Corequisites

- Prerequisite: Accepted in an engineering graduate program.

3 Catalog Course Description

Fundamentals of discrete-time signal processing; data acquisition; analog-to-digital and digital-to-analog conversions; sample rates; aliasing and anti-aliasing filtering; spectral analysis and system identification; discrete-time filter design and implementations.

4 Learning Outcomes

Students will develop knowledge of the concepts and theory of digital signal processing and increase their understanding of frequency domain analysis, which can be used for various topics such as filter design and spectral analysis.

After taking this course, students will be able to do the following:

1. Analyze discrete-time signals in both time- and frequency-domains.
2. Compute the Discrete Fourier Transform of discrete-time signals.
3. Find system transfer function based on the z-transform.
4. Relate system transfer function and frequency-domain response of a linear system.
5. Design digital filters based on required frequency response and stability.
6. Simulate discrete-time systems using software tools.

5 Required Textbook and Class notes

- No required textbook.
- Class notes will be posted online.

6 Recommended Books

- Alan V. Oppenheim and Ronald W. Schaffer, *Discrete-Time Signal Processing*, Third Edition, Prentice Hall, 2009.

7 Required Hardware

- A laptop computer running Windows, macOS, or Linux.
- For Windows users, we highly recommend you to use WSL. Instructions will be provided.

8 Assessment of Learning Outcomes

8.1 General Guidelines

- There will be various in-class activities to help students learn well in class.
- There will be bi-weekly homework assignments, which may include programming assignments.
- There will be bi-weekly quizzes on Fridays.
- There will be a mandatory final exam.
- There will be a course project.

8.2 In-Class Activities

In-class activities include:

- Unannounced Polleverywhere feedback and reading quizzes.
- Board work for solving class problems.

8.3 Homework Assignments

HW assignments will not take up too much time if you truly understand the topics.

You are encouraged to discuss HW problems with your peers, but you cannot copy solutions from them. You have to make sure you understand everything you write on the answer sheet.

8.4 Bi-Weekly Quizzes

Bi-weekly quizzes will mainly cover the content that is covered by the most recent submitted homework problems.

For Quiz n , you can use n pieces of letter-sized paper for formula sheet, which can be double-sided.

8.5 Final Exam

The final exam is mandatory. The location is the same classroom we use.

- Formula sheet:
 - No notes, computer, or smart phone can be used.
 - You can use 7 pieces of letter-sized paper for formula sheet, which can be double-sided.

- Coverage:
 - The difficulty level will be similar to those of the bi-weekly exercises/quizzes.
 - Problems may include everything covered in the entire semester, including HW 13.
 - The amount work will be about 2.5 times of a normal bi-weekly exercises/quizzes. Exercises/quizzes are designed for 40 minutes and the final for 120 minutes.

8.6 Class Project

According to federal and university policies, a full-time status of undergraduate and graduate students require 12 credits and 6 credits, respectively. As such, we know that the requirements for undergraduate and graduate courses are different; the latter have much higher workload.

For CEC 510, we will have a class project, which should be about 30% of the total workload of the class. Here are the basic guidelines for the class project:

- The topic will be about applications of audio signal processing. A list of projects will be provided in the class.
- Students can form teams of TWO students to perform the project. Each student will be given half credits unless there is a specific indication in the final report.
- There are bi-weekly progress reports, aligned with the submission of the first 6 homework.
- There is a submission of final report, aligned with the submission of HW 7.
- There will be a class presentation; the allocated time for each team will be announced later.

8.7 Submission of Work

For all the submissions, students are expected to act in a professional manner, including:

- Submitting work on time. Assignments will be given enough time to finish. **No late submission is accepted.**
- Active, collaborative learning is proven effective for improved learning. Students are encouraged to form teams of 2 to 3 students to discuss ideas of homework problems and projects in and out of class. Yet, each student should write their solutions independently.
- All work turned in for grading must be done neatly and professionally on Letter-sized paper, leaving 1-inch margins for making copy and binding. For electronic submission, please submit PDF only except otherwise specified. Use correct approach and show all necessary intermediate steps. Draw neat, clearly labeled sketches as necessary; all the variables used must be clearly defined.
- Typing program code neatly following the 1TBS (one true brace style) format (with the 4-space indentation), as illustrated below.

```
1  int abs(int x) {
2      if (x >= 0) {
3          return x;
4      } else {
5          return -x;
6      }
7  }
```

9 Class Schedule

Below, we only list the Lecture number in the course schedule. For details of contents of each lecture, see the content map at the end of this document.

Date and Day	Lecture	Notes
8/24/26, Monday	Intro 1	Intro to the Class/ graduate projects
8/26/26, Wednesday	Lctr 1	Intro to graduate projects
8/28/26, Friday	Lctr 2	
8/31/26, Monday	Lctr 3	HW 1 released
9/2/26, Wednesday	Lctr 4	
9/4/26, Friday	Lctr 5	
9/7/26, Monday	No class	Holiday—Labor Day
9/9/26, Wednesday	Lctr 6	HW 1 due; HW 2 released
9/11/26, Friday	Lctr 7	
9/14/26, Monday	Lctr 8	
9/16/26, Wednesday	Quiz 1	Early alert: effort-based
9/18/26, Friday	Lctr 9	HW 2 due; HW 3 released
9/21/26, Monday	Lctr 10	
9/23/26, Wednesday	Lctr 11	
9/25/26, Friday	Lctr 12	HW 3 due; HW 4 released
9/28/26, Monday	Lctr 13	
9/30/26, Wednesday	Lctr 14	
10/2/26, Friday	Quiz 2	
10/5/26, Monday	Lctr 15	HW 4 due; HW 5 released
10/7/26, Wednesday	Lctr 16	
10/9/26, Friday	Lctr 17	
10/12/26, Monday	Lctr 18	HW 5 due; HW 6 released
10/14/26, Wednesday	Lctr 19	Mid-term grade: performance-based
10/16/26, Friday	No class	Fall Break
10/19/26, Monday	Lctr 20	
10/21/26, Wednesday	Quiz 3	
10/23/26, Friday	Lctr 21	HW 6 due; HW 7 released
10/26/26, Monday	Lctr 22	

Date and Day	Lecture	Notes
10/28/26, Wednesday	Lctr 23	
10/30/26, Friday	Lctr 24	HW 7 due; HW 8 released
11/2/26, Monday	Lctr 25	
11/4/26, Wednesday	Lctr 26	
11/6/26, Friday	Quiz 4	
11/9/26, Monday	Lctr 27	HW 8 due; HW 9 released
11/11/26, Wednesday	Lctr 28	
11/13/26, Friday	Lctr 29	
11/16/26, Monday	Lctr 30	HW 9 due; HW 10 released
11/18/26, Wednesday	Lctr 31	
11/20/26, Friday	Lctr 32	
11/23/26, Monday	Quiz 5	HW 6 due
11/25/26, Wednesday		Holiday
11/27/26, Friday		Holiday
11/30/26, Monday	Lctr 33	HW 10 due
12/2/26, Wednesday	Presentations	Graduate proj presentations
12/5/26, Saturday	Final exam	5:00 pm to 7:00 pm

10 Grading Weighting and Letter Grade Assignment

Your class grade is the weighted average of various assessments.

Assessment	Points	Assessment	Points
In-class activities	2 points	HW	15 points
Bi-weekly quizzes	25 points	Final exam	25 points
Class project report	30 pts	Project presentation	3 points

Total points: 100

Grading Scale: 90+ => A, 80+ => B, 70+ => C, 60+ => D, 60- => F.

11 Class Policies

11.1 AI Use Policy for This Course

Artificial Intelligence (AI) tools can be valuable for learning programming when used thoughtfully. To help you benefit from these tools while ensuring you develop your own programming skills, the following policies apply:

- **Permitted Use**

- You are encouraged to use AI tools (e.g., ChatGPT, Copilot, Augment Code, Roo Code, Msty, Ollama, etc.) to help understand course content and to support your work on assignments.
- If you use AI tools for your assignment, you should clearly indicate it.

- **Not-permitted Use**

- AI tools CANNOT be used in quizzes and the final exam.

- **Accountability for Understanding**

- Regardless of whether your code is handwritten or generated with AI assistance, **you must be able to explain your code when asked** (in class, during labs, or on assessments).
- If you cannot explain how your code works, you will not receive credit for that portion of the assignment.

- **Learning First, AI Second**

- You are encouraged to **first attempt writing code by hand**, and then use AI tools to explore alternative approaches. This helps you build your own problem-solving ability before comparing with AI-generated suggestions.

- **Scope of Knowledge**

- AI-generated code should primarily use the concepts and techniques we have covered in class.
- TBD If your AI-assisted solution includes ideas or libraries **beyond the current course content**, you must provide a brief explanation of why you chose that approach and how it connects to the problem at hand.

- **Purpose of AI in This Course**

- The goal of using AI tools is not just to finish assignments faster, but to **reinforce your understanding of programming concepts** and to give you opportunities to compare, contrast, and reflect on different solutions.

11.2 Classroom Policies

- In the class, no electronic distractions, such as texting or web browsing, will be allowed.
- Small talks that are not related to the subject of the practices is not allowed in the classroom. Repeated violations can result in the dropping of the student from the class.
- Students are encouraged to ask questions when they have.

- If the questions are only specific to very few students, these students are invited to visit the instructor during the office hours.

11.3 Missing Assignments

- No makeups for missing bi-weekly HW as they can be submitted online.
- No makeups for missing quizzes. Yet, we only count the best 4 quizzes so missing one is not a big deal.

11.4 Academic Integrity

ERAU is committed to intellectual integrity and considers academic dishonesty a very serious offense. Such offenses include cheating (accepting or giving unauthorized assistance in preparing assignments), or plagiarism (taking the ideas, writings, and/or work of another and representing them as one's own without appropriate acknowledgment). A student who cheats or commits any form of academic fraud will be subject to a failing grade for the course. In addition, the incident will be reported to the Dean of Students. If any other academic integrity violations have been documented, the student will be recommended for dismissal.